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An investigation of prospective English language teachers' multimodal literacy

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Abstract

With the development of digital communication technologies, we are required to re-define the term *literacy* as *"multimodal literacy"*. The current generations, as digital natives, are literate not only in the conventional sense but in an innovative way which involves texts with static and dynamic images, gestures, audio, spoken language and written language. Citizens of today need to be able to comprehend meaning that is constructed through two or more communication modes. Likewise, it is equally important that teachers of the digital world need to possess multimodal literacy knowledge and skills. The teachers should be able to comprehend and interpret multimodal texts, and effectively design and communicate meaning through such texts and finally need to transfer this knowledge and skills to their students. This quantitative study aims to investigate multimodal literacy levels of 383 pre-service English language teachers at Gazi University, Turkey. The multimodal literacy of prospective teachers are assessed through the Multimodal Literacy Scale developed by Bulut, Ulu and Kan (2014) and the scores are evaluated in terms of gender, grade, and other background features such as internet use and parents' education level. Conclusions are drawn and suggestions are made for the English language teaching programs.

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1. Introduction

The last few decades have witnessed emergence of new technologies and a diverse array of media we communicate. Today the way we live and communicate often includes print along with use of email, texting,

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voicemail, and Internet web-pages with image, sound, and music. Eventually, the notion of "text" has broadened to include digitally constructed texts constituted through more than one mode or means of representation such as words, dynamic and static images, sounds, symbols and gestures. These texts are dynamic, nonlinear and interactive. To be able to function in today's world, a shift is required in literacy training from the conventional sense towards producing and understanding texts involving different modes of human communication such as visual, verbal and gestural (Leu, 2002; Harste, Woodward & Burke 1984: 208). Hence multimodal literacy, defined as "a framework that requires a collective interpretation of two or more scripts, visuals, videos, graphics, animations, sounds, music, gestures and facial expressions for producing meaning" (Kress, 2010: 54). Multimodal texts are inherently associated with digital technologies because in many digital texts, different modalities, aural, visual, gestural, spatial and linguistic come together to construct meaning. Multimodal literacy is about understanding the different ways of knowledge representations and meaning-making, understanding discourse by investigating the contributions of specific semiotic resources such as language, gestures and images, using various modalities such as visual and aural elements to create meaning and understanding how various modalities co-work to construct a coherent text.

Digital natives surrounded and shaped by these technologies act differently in communication and meaning making. Literacy training based on words on a piece of paper ignores other ways of communication and interaction and is insufficient to train and educate students for life and work in the digital world (Alvermann, 2002; Buckingham, 2003; Cope & Kalantzis, 2000; Gee, 2003; Jewitt & Kress, 2003; Kress, 2003; Lankshear & Knobel, 2003; O'Brien & Bauer, 2005). Given the fact that more and more students arrive at school more competent in new technologies and literacies than their teachers (Chandler-Olcott & Mahar, 2003; Leu, Kinzer, Coiro & Cammack, 2004), changes need to be done in education and teacher training.

Ignoring this phenomenon in our classrooms would be a mistake. If we do so, we run the risk of losing touch and school may become boring and irrelevant for students as a result. The gap between conventional print-based schooling and the new multimodal literacy practices is called "the digital divide and disconnect" (O'Brien & Bauer, 2005: 126). Students are more interested and competent in new literacies than their teachers who experience problems creating and understanding multimodal texts using new technologies (O'Brien & Bauer, 2005, Jewitt et al., 2007). Print-only instruction is far from addressing to the needs and interests of students of the digital world and lack of multimodal literacy practices is often cited as resulting in increasing student disengagement (Gee 2004). Hence, it is recommended that educators should hold a more constructive approach to benefit from new technologies at play on students' literacy practices.

The current generations, as digital natives, are literate not only in the conventional sense but in an innovative way which involves texts with static and dynamic images, gestures, audio, spoken language and written language. It is equally important that teachers of the digital world also need to possess multimodal literacy knowledge and skills. The teachers should be able to comprehend and interpret multimodal texts, and effectively design and communicate meaning through such texts and finally need to transfer this knowledge and skills to their students.

Research indicates that using multimodal texts contributes to students' use of cognitive skills and increases their creativity, attention, participation and production (Callow & Zammit, 2012) .Walsh (2010: 220) suggests that the use of multimodal texts in language skills courses changes the nature of reading, writing, speaking and listening skills. The use of multimodal texts not only changes in the process of teaching a class, testing and evaluation, participation in courses but also the roles of teachers and learners (Walsh 2010: 222). Introduction of hardware, software and the Internet does not guarantee technology integration to classrooms. The teachers need opportunities to make use of multimodal literacies in their own lives before integrating them effectively to their teaching practices (Lankshear & Knobel, 2003). To this end, teacher training programs should involve knowledge and skills to integrate multimodal literacies to their practices.

As to learning a foreign or second language, teachers may often be resistant to integrate multimodal practices to their teaching (Tan & Mc William, 2009; Warschauer, 2008). "Reducing L2 learning to the flat literacies of paper-based resources in the classroom raises questions of authenticity in L2 learning" (Lotherington & Jenson 2011: 228). To design authentic communicative activities necessitates use of new dimensions of literacy as they are in indispensable part of life and communication today. What is more, use of multimodal texts makes it possible to draw on students' out-of-school literacies to help them perform better in our classrooms (Sewell & Denton 2011: 61).

1.1. Statement of the problem

In the Turkish education context, the need for teachers who has multimodal literacy is evident. FATIH Project, which is a national project in piloting phase now, involves equipping all classrooms with smartboards, Internet connection and digital learning materials along with providing tablets for all students. Conducting instruction in these classrooms definitely requires multimodal literacy skills for the teachers. Hence, this study aims to investigate the multimodal literacy levels of pre-service English language teachers.

2. Methodology

2.1. Design

This quantitative study aims to investigate multimodal literacy levels of pre-service English language teachers. A questionnaire consisting of items related to demographics and practices of participants along with the Multimodal Literacy Scale developed by Bulut, Ulu and Kan (2014) is used to collect data. The Multimodal Literacy Scale contains 17 items with three 3 factors: (1) Expressing Oneself Using Multimodal Structure; (2) Interpretation of the Contents Presented in Multimodal Structure and (3) Preferring Multimodal Structures. The participants indicated their agreement about the statements on a five-point Likert scale.

The participants are 383 pre-service teachers attending English language teacher education program at Gazi University, Turkey. The program is a four-year training program and an additional foundation year at the beginning for intensive language instruction.

2.2. Data Analysis

The statements in the Multimodal Literacy Scale were assessed with descriptive statistics about mean scores and standard deviation and to what extent the pre-service teachers agreed with the items were found. Each statement in the scale was processed with single sample T test. As to gender, personal computer and use of social media, independent samples T test was used. One-way ANOVA test was used to investigate year, parents' education and daily time on the net. When there was a significant difference, Post-Hoc Tukey and LSD tests were used to find out the difference between groups.

3. Findings and Discussion

The table below shows demographics of the participants with reference to gender, year at the training program, parents' education, whether or not they possess a personal computer, the time they spend daily on the internet and their use of social media.

Table 1. The demographics of the participants

		N	%
gender	female	323	84,3
	male	60	15,7
Year	Foundation year	71	18,5
	1	78	20,4
	2	78	20,4
	3	93	24,3
	4	63	16,4
Mothers' education	primary	197	51,4
	secondary	63	16,4

	high	84	21,9
	university	36	9,4
	MA	3	0,8
Fathers' education	primary	92	24
	secondary	64	16,7
	high	127	33,2
	university	90	23,5
	MA	9	2,3
	Ph. D	1	0,3
Personal computer	Yes	354	92,4
	No	29	7,6
Time on the Internet	Less than an hour	58	15,1
	1-3 hours	187	48,8
	3-5 hours	113	29,5
	More than 5 hours	25	6,5
Use of social media	Yes	361	94,3
	No	22	5,7

The table below shows descriptive statistics related to the mean scores and standard deviations expressing agreement levels of the pre-service teachers for each of the statements in the scale. To calculate the score intervals, "Range width= Series width/ number of groups" (Tekin, 2000). It was determined that scores in the interval of 4,20-5,00 show total agreement, 3,40-4,19 show agreement, 2,60-3,39 show undecided, 1,80-2,59 show disagreement and finally 1-1,79 show total disagreement.

Table 2. Mean scores of statements in the Multimodal Literacy Scale.

Multi modal Literacy Skills	statements	mean	St dev.	t
Expressing Oneself Using Multimodal Structure	Using various elements (such as music and images) in my presentations makes it easier to make my point.	4,2585	,86134	96,756
	I use visuals such as graphics, tables, pictures and photographs in my writings.	4,2428	,82566	100,566
	I prepare an interactive presentation making use of music, visuals and animations.	4,1619	,85303	95,483
	I organize my thoughts systematically in my presentations thanks to various visual elements (such as tables and graphics).	4,0731	,82809	96,260
	I express myself more explicitly in environments in which writing, sound and images exist together.	4,0601	,87373	90,940
Interpretation of the Contents Presented in Multimodal Structure	I pay attention to the body language of the individuals I am listening to.	4,5091	,66655	132,392
	I can realize how visual, auditory and written elements influence individuals.	4,2533	,75964	109,576
	I use body language which is in harmony with the words I choose when speaking.	4,1358	,73957	109,440
	I relate various visual and verbal information on various media tools to each other.	4,1149	,70329	114,504
	I interpret the information that I gather from numerous resources.	4,1097	,79822	100,759
	I relate the information I have access to each other using visual and auditory elements.	4,0783	,79536	100,351
	I can decide whether contents that are presented on various media (newspaper, TV,	3,8277	,81963	91,394

	social media etc.) are true or not.			
Preferring Multimodal Structures	I don't like trying to interpret images, sounds, graphics and writings simultaneously.	2,4178	1,17918	40,126
	I only believe in the power of verbal expressions when sharing my thoughts.	2,2298	1,13232	38,538
	I get distracted in electronic environments in which visual, auditory and written elements are used together.	2,2219	1,12345	38,706
	The use of visual, auditory and written elements together leads to laziness of the mind.	2,1279	1,06924	38,948
	I get bored in communication in which written, auditory and visual elements are used together.	2,0104	1,07562	36,579

Table 3 displays multimodal literacy levels of the pre-service teachers in terms of the three subscales. As can be seen, the scores indicate agreement in *expressing oneself using multimodal structure* and *interpretation of the Contents Presented in Multimodal Structure*. The third factor *preferring multimodal structures* received the lowest scores. Yet, as the statements were negatively constructed, lower scores in this subscale indicate importance to and preference for multimodal literacy structures. Pre-service teachers appear to be able to use, interpret and prefer multimodal structures at high levels.

Table 3. Multimodal literacy levels of pre-service teachers in terms of subscales

Multimodal literacy skills	mean	St dev	t
Expressing Oneself Using Multimodal Structure	4,1593	,59861	135,979
Interpretation of the Contents Presented in Multimodal Structure	4,1470	,46851	173,224
Preferring Multimodal Structures	2,2016	,80632	53,435

Below are multimodal literacy sub-scale scores with reference to gender. Two of the subscales indicate significant difference with respect to gender. A closer look at the items shows that the differences are "I prepare an interactive presentation making use of music, visuals and animations." (,036*), "I interpret the information that I gather from numerous resources" (,006**), "Using various elements (such as music and images) in my presentations makes it easier to make my point" (,001**) and "I relate the information I have access to each other using visual and auditory elements" (,015*); all of which favoring female pre-service teachers.

Table 4. Multimodal literacy sub-scales with reference to gender.

Multimodal literacy skills	Gender	Mean	St dev.	t	p
Expressing Oneself Using Multimodal Structure	Female	4,1944	,56621	2,689	,007**
	Male	3,9700	,72610		
Interpretation of the Contents Presented in Multimodal Structure	Female	4,1716	,45273	2,156	,034*
	Male	4,0143	,53043		
Preferring Multimodal Structures	Female	2,1814	,80040	-1,101	,274
	Male	2,3100	,83599		

*Independent-Samples T-Test *p<0,05 ; **p<0,01*

Table 5 below shows multimodal literacy skills in three subscales with respect to year. It appears that there is significant difference in two subscales: expressing oneself using multimodal structure and preferring multimodal structures. The difference is between the first year and the rest. The reason for the difference might result from the requirements of the courses taken in the years. The pre-service teachers take skills courses and lectures in the first year in which they are not required to do presentations or prepare materials. However, the other years, especially in the third and fourth years, the pre-service teachers take methodology courses. In all of these courses they do

microteaching demonstrations, prepare lesson plans and materials. As they widely use multimodal structures to perform these, there appears to be a significant difference in expressing themselves using multimodal structures and preferring them.

Table 5. Multimodal literacy subscale scores in terms of year.

Multimodal literacy skills	N	Year	mean	St d	F	p	sig
Expressing Oneself Using Multimodal Structure	71	F. year	4,1606	,52221			1-2
	78	1	3,9179	,69014			1-4
	78	2	4,3256	,48895	5,728	,000**	1-3
	93	3	4,1333	,60816			
	63	4	4,2889	,58008			
Interpretation of the Contents Presented in Multimodal Structure	71	F. year	4,1087	,45011			
	78	1	4,0256	,50847			
	78	2	4,1996	,39298	2,175	0,071	
	93	3	4,1997	,48130			
	63	4	4,1973	,48644			
Preferring Multimodal Structures	71	F. year	2,0930	,60741			H-3
	78	1	2,2128	,71883			1-3
	78	2	2,1460	,74414	3,904	,004**	2-3
	93	3	2,4624	1,02998			4-3
	63	4	2,0231	,72575			

*One Way-ANOVA; LSD Test **p<0,01*

The table below shows multimodal literacy subscale scores with respect to the time on the net daily. The findings suggest a significant difference in *expressing oneself using multimodal structures* between those who use it less than an hour and the rest. It appears that multimodal literacy levels increase in line with time spent on the net.

Table 6. Multimodal literacy subscale scores in terms of time spent on the Internet daily

Multimodal literacy skills	N	Year	Mean	St deviation	F	p	
Expressing Oneself Using Multimodal Structure	58	Less than 1 hour	3,9793	,72421			1-2
	187	1-3	4,1572	,55677	2,768	,042*	1-3
	113	3-5	4,2195	,59219			1-4
	25	More than 5	4,3200	,54467			
Interpretation of the Contents Presented in Multimodal Structure	58	Less than 1 hour	4,0369	,55976			
	187	1-3	4,1490	,42596	1,480	,219	
	113	3-5	4,1871	,45017			
	25	More than 5	4,2057	,59773			
Preferring Multimodal Structures	58	Less than 1 hour	2,2931	,85118	,308	,820	
	187	1-3	2,1936	,72732			

	113	3-5	2,1735	,91710	
	25	More than 5	2,1760	,76009	

*One Way-ANOVA; LSD Test *p<0,05*

As final findings of the study, the analyses do not indicate any statistical difference with respect to parents' education, socio-economic status, their program (day-evening), or possessing a personal computer or not.

4. Conclusion and Suggestions

Today, teachers have the opportunity to create rich learning environments for their students, enhanced by the immense information and resources on the Internet by using different modes. English teachers need support to learn about and be able to use new kinds of multimodal learning in order to be prepared for teaching students in 21st century classrooms. The findings of this study suggest that pre-service English language teachers have quite high multimodal literacy levels. The skills and knowledge increases with time spend on the net, year of study and gender. The more time is spent on the net and the more multimodal structures are used as course requirements, the more multimodal literate pre-service teachers become. This finding can be transferred to other language teacher education institutions but for wider generalizations, replica studies are needed. One important suggestion for further study is to assess multimodal literacies of practicing teachers with respect to experience and graduation. Practising teachers, who are at various ages, might display different profiles in multimodal literacy. For the successful implementation of rich learning environments, all teachers should have an optimal level of multimodal literacy.

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